

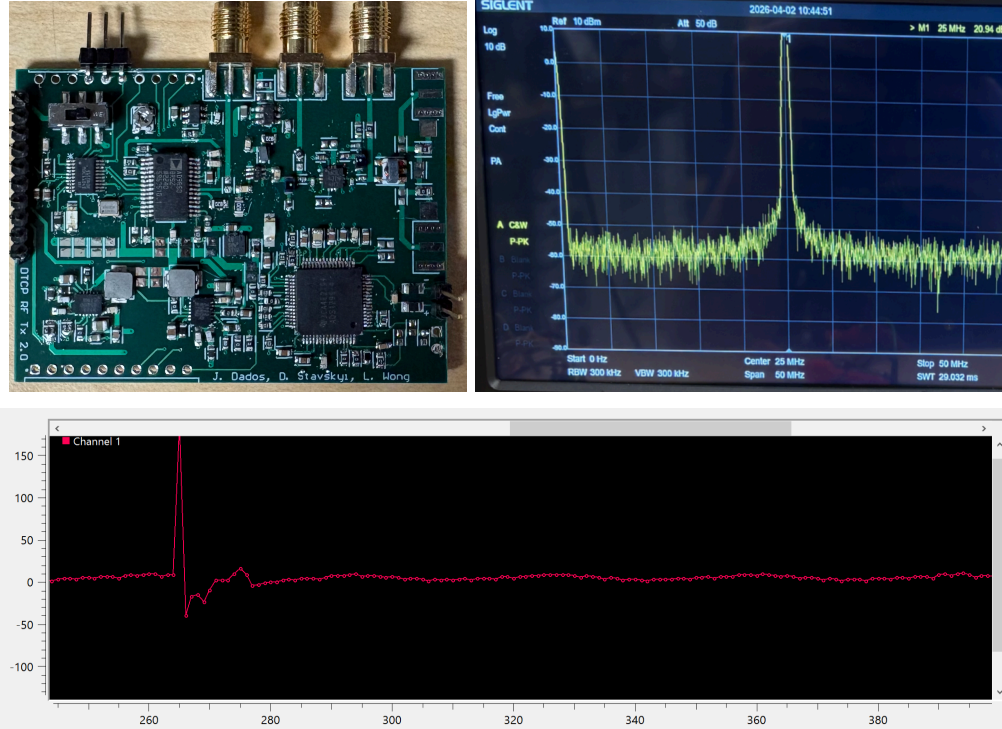
Production Release

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Project repository: <https://github.com/jdados/dtcp>

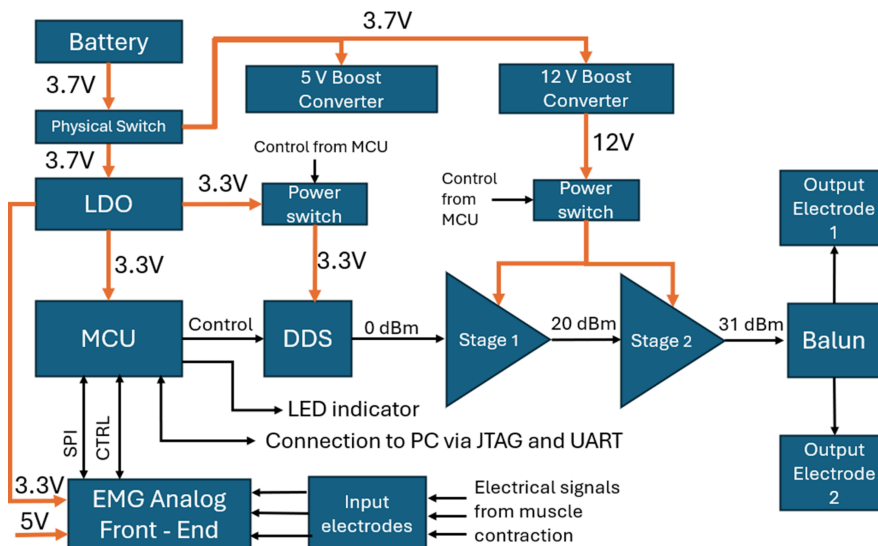
1. Current state of the project work

The final device is assembled and has been tested using the validation setups developed for the previous builds:



The production release device, its RF output power, and an EMG signal captured with it

The initial testing of the device revealed the need to reduce the drain voltages of the power amplifiers from 13V to 12V for increased stability without losing any output power. We were able to accomplish that on the same PCB layout as before, because the only change needed was replacing the voltage divider resistors on the buck converter. Hence, the updated system architecture for the device is shown below:



Updated system architecture with new supply voltage for the PAs

The testing in rats was delayed due to surgery scheduling conflicts with other projects at the lab. These experiments will be completed next week and the device will be shipped to Harvard afterwards.

2. Usability

a. Interface

The UART plotting interface of the device is fully operational and bug free. We are working on making a device manual for researchers to operate it without needing a demonstration from us.

b. Navigation and perception

The device needs to be connected to a battery and to a PC through 2 UART wires. After that, the user simply needs to move the power switch on the board to the “on” position and set the output power of the RF transmitter using the potentiometer.

c. Responsiveness

The power switch and power potentiometer provide physical ways of controlling the electronic device by the user. There are also LEDs to indicate power on and active RF stimulation.

3. Build quality

a. Robustness and Consistency

All major functionalities have been fully implemented and validated to ensure no bugs. The board operates bug-free over tests spanning several hours.

b. Aesthetic rigor

The board layout was made following the aesthetics of production-level PCBs. That involved ensuring short signal paths and logical placement of the components on the board.

4. Vertical features

a. External interface

All the external interfaces are verified and carry out functionality correctly. The previous synchronization issues in the UART interface have been resolved.

b. Persistent State

The UART interface enables persistent data storage through the SerialPlot functionality of saving a CSV file of the data stream. Hence, the only persistent state of interest in the project has been implemented successfully.

c. Internal Systems

The PCB design of the Production Release captures all learnings and bug fixes from the candidate version. The communications between all internal systems (such as SPI between MCU and AFE) have been redesigned in hardware to eliminate any issues.